Dual Causal-Aware Detection Transformer for Remote Sensing Images

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***Abstract*—Deep neural networks often inherit biases from training data, compromising generalization. In visual recognition, distinguishing foreground from background is insufficient, as models tend to rely on spurious correlations rather than learning essential causal patterns. To address this issue, this article proposes a novel transformer architecture, termed dual causal-aware detection transformer (DCDT), specifically designed for object detection in optical remote sensing images from a causal perspective. Specifically, we begin by constructing a structural causal model (SCM) to intuitively analyze the causal effects inherent in the overall visual patterns. Building on this foundation, DCDT introduces causal constraints at the attention level by embedding dynamic multiscale causal prototypes into the attention mechanism. The derived causal priors are subsequently used to enhance features at the representation level, thereby enforcing feature-level causal modulation. This dual causal-aware strategy enables the precise extraction and reinforcement of causally relevant features, improving both robustness and discriminative capability in complex detection scenarios. In addition, a sparse kernel-region mask is incorporated to decouple local information from global representations, effectively strengthening the modeling of fine-grained structures. Extensive experiments conducted on two challenging public datasets, DIOR and HRRSD, demonstrate that DCDT consistently outperforms existing methods and baselines. These results validate the effectiveness of DCDT in capturing both global causal semantics and local fine-grained features, highlighting its practicality in complex remote sensing scenarios.**

***Index Terms*—Causal prior, dual causal-aware strategy, object detection, optical remote sensing images.**

I. Introduction

THE data-driven nature of deep neural networks causes biases in the training data to directly affect the generalization ability of the model [1], [2]. To mitigate data bias, researchers have proposed a variety of compensation methods, such as reducing potential bias through adversarial training [3], [4], aligning feature distributions using domain adaptation techniques [5], employing attentional enhancement [6], [7], [8] and feature denoising [9], [10], [11] to identify key features and minimize the impact of bias, and designing bias-aware loss functions to perform sample reweighting [12], [13], among others.

Despite these significant advances, existing approaches focus primarily on feature extraction and optimization strategies, failing to distinguish between causal and noncausal components within features. Moreover, they often neglect the inclusion of domain-specific a priori knowledge [14], [15]—a critical limitation, as such knowledge can provide contextual cues for object detection in complex scenes. To address these challenges, we introduce a causal modeling-based approach [16], [17] that aims to disentangle true causal factors from spurious correlations in observational data. Unlike purely data-driven methods, causal models are designed to capture the structural relationships among variables, thereby enabling the identification of causally relevant features that are robust to distribution shifts and domain-specific biases. By leveraging causal inference, it becomes possible to incorporate domain priors and intervene on latent mechanisms during training or inference, which enhances both the generalization ability and interpretability of the model. In our work, causal modeling is integrated into the feature learning pipeline to guide the model in emphasizing features with genuine predictive power for object recognition tasks.

It is worth noting that both foreground and background features contain a mixture of beneficial and harmful biases [18], [19]. For instance, ships in harbors typically appear in watery regions, and this spatial bias can aid detection. Conversely, background textures (e.g., buildings near vehicles) may introduce confounding correlations. Similarly, within objects, geometric structures like vehicle wheels are causally associated with the target class, whereas over-reliance on lighting-sensitive features (e.g., skin tone in face recognition) introduces harmful, noncausal dependencies. Previous approaches have attempted to mitigate bias by decoupling foreground and background [20], [21]. However, such strategies often overlook the underlying causal relationships embedded within.

In this article, we focus on constructing a robust recognition model that addresses the bias problem through the

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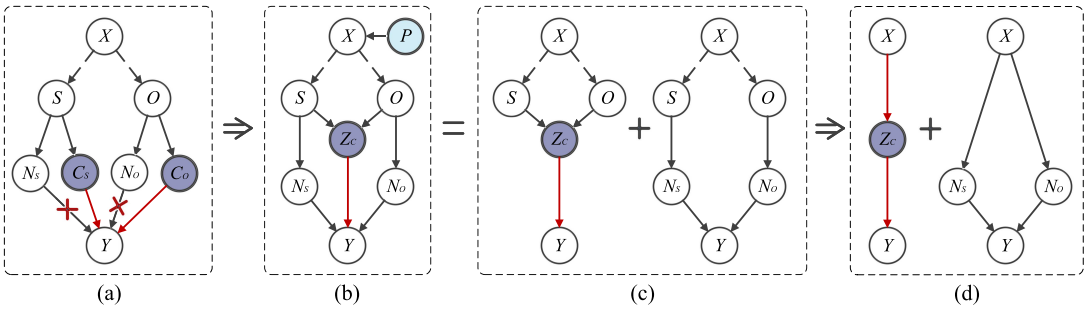


Fig. 1. Illustration of the decomposable structural causal models. (a) Intuitively, image features are divided into foreground and background, each further decomposed into causal and noncausal components. (b) Causal constraints are introduced to extract causal effects. (c) Separate causal effect paths from confounding effect paths. (d) Final causal pathway is adopted for causal effect computation without explicit foreground–background separation.

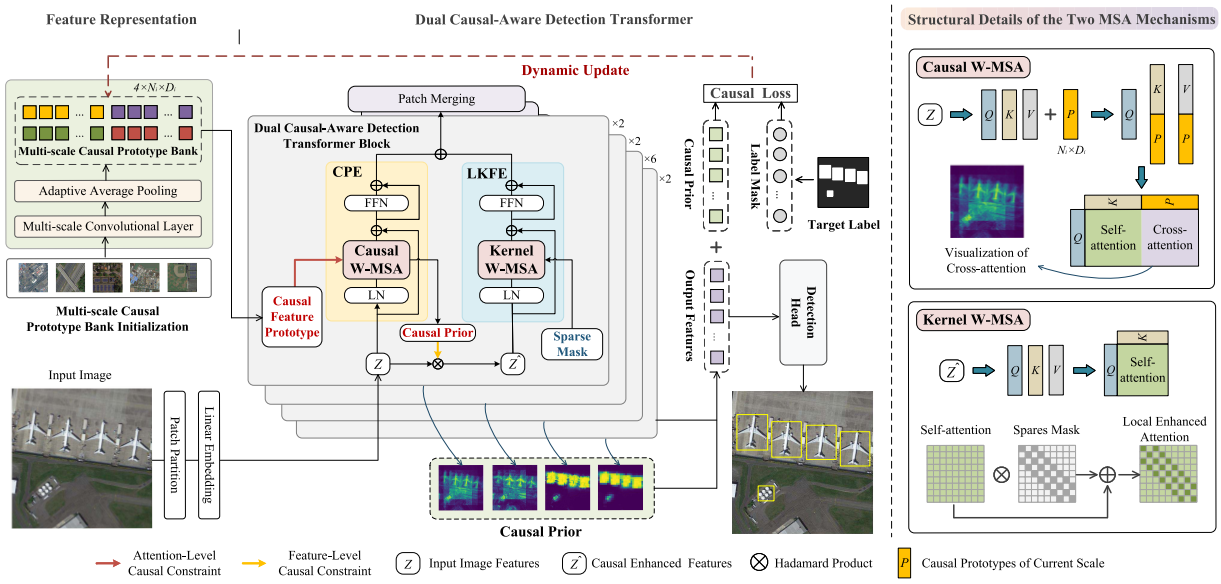


Fig. 2. Architecture of DCDT. This includes the construction of the DCDT block, where each block is composed of a CPE module and an LKFE module. The main mechanisms of the two attention modules, Causal W-MSA and Kernel W-MSA, are also presented.

introduction of causal learning [22], [23]. Building on the causal nature of visual recognition systems, we develop structural causal models (SCMs) [24] to analyze the causal relationships among image features, data bias, and prediction results as illustrated in Fig. 1. Intuitively, the image feature X comprises both foreground information (O) and background information (S). However, rather than performing a simple foreground-background separation, we further decouple both components into causal features (C_S, C_O) and noncausal features (N_S, N_O), as illustrated in Fig. 1(a). This model unveils the underlying causal structure of visual recognition tasks, wherein the prediction label Y is determined by both causal and noncausal features derived from the image features X. Building upon this foundation, we further consolidate the causal features into a unified representation Z_C, isolating it from the overall feature space, as depicted in Fig. 1(b) and (c). The goal is to treat causal features as an integrated entity, thereby facilitating a comprehensive examination of their role in visual recognition.

To enhance the role of causal features and eliminate the interference caused by spurious correlation paths, this article proposes a dual causal-aware detection transformer (DCDT) for object detection in remote sensing images. The model architecture is shown in Fig. 2. It consists of two main components: causal prior extraction (CPE) and local kernel-region feature enhancement (LKFE).

The CPE module is based on the theoretical framework of SCMs and incorporates dual causal enhancement strategies into the transformer architecture. Causal constraints are introduced at the attention-level by embedding multiscale causal prototype bank into the transformer attention mechanism, guiding the model to focus on causal features that are directly relevant to object recognition. Subsequently, cross-attention between image features and causal prototypes is computed during training to derive a causal prior, which is then used to impose a feature-level constraint on image features. This ensures that only features highly correlated with the causal prototypes are selectively enhanced. Through contrastive learning optimization, the prototype features are

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optimized iteratively during the training process, with their representations asymptotically approaching the true causal features.

The LKFE module introduces a novel kernel-region attention mechanism that enhances fine-grained local features by decoupling local information from broader attention patterns. This is achieved through a sparse mask (SpaM) matrix that selectively preserves connections within contiguous kernel blocks while suppressing irrelevant long-range responses. This design not only amplifies local patterns to improve sensitivity to contextual details but also reduces background noise around key object regions. Theoretically, when explicit sparsity constraints are imposed, the attention mechanism can be interpreted as a dynamically generated graph adjacency matrix. The applied sparsity constraints resemble neighborhood aggregation in graph convolutional networks, maintaining both local smoothness and global structural consistency [25], [26]. This conceptual framework provides theoretical justification for our approach. Furthermore, from a causal perspective, this selective subgraph construction places greater emphasis on local structures that carry direct causal signals, thereby reinforcing meaningful information flow.

Ultimately, the entire system forms a complete causal learning framework: it enables causal attention modulation based on a multiscale causal prototype bank and performs feature space interventions guided using causal priors. Experimental results on complex remote sensing datasets demonstrate that this causality-aware architecture significantly boosts detection accuracy in challenging scenes.

Overall, our contributions are summarized as follows.

1) DCDT is designed as an object detection network for optical remote sensing images, enabling accurate multiscale object detection in challenging and complex background environments.
2) A dual causal-aware strategy is proposed to impose causal constraints at both the attention and feature levels, enhancing the model's ability to focus on task-relevant causal information.
3) LKFE module based on causal feature inputs is developed, which decouples the local attention of the kernel region from the attention map of the current window features by constructing a SpaM matrix.

The rest of this article is organized as follows. In Section II, we provide a brief review of related works. Section III presents a detailed introduction to the proposed DCDT. Section IV describes the experimental setup and the analysis of the results, and performs ablation studies of the proposed model. Finally, the conclusion is drawn in Section V.

II. Related Works

A. Transformer Detector for Remote Sensing Images

Researchers have increasingly adopted transformer architectures for object detection tasks in remote sensing imagery to improve the performance of the classical transformer framework [27], [28]. Zeng et al. [29] enhanced the DETR model by incorporating a rotationally deformable attention mechanism, embedding angular information directly into the detector. Hu et al. [30] improved DETR by incorporating ignored samples into the training process, enhancing model training efficiency. Guo et al. [31] proposed a cross-scaling mechanism to resolve foreground–background confusion through a local cross-attention mechanism. These studies refined the transformer's attention mechanisms [32], [33] to retain its capability for extracting long-range dependencies while incorporating localized or adaptive attention methods to better capture fine-grained details. Their global receptive field, although powerful, can dilute fine-grained spatial cues crucial for detecting small or occluded objects.

Further combining the strengths of CNNs and transformers, Xu et al. [34] extended the Swin Transformer by introducing a local perception module and inflating convolutional operations, enhancing the model's ability to extract localized features. Xu et al. [35] aligns the semantic information extracted from the transformer branch into a single-branch CNN during training. These advancements underscore the synergy between CNN and transformer-based approaches [36], [37], optimizing their application to complex remote sensing tasks. But these hybrids still lack mechanisms to explicitly disentangle causal features from spurious correlations in remote sensing data.

B. Causal-Aware Strategy in Deep Learning

Causal inference [38] has emerged as a significant paradigm within deep learning, aiming to improve generalization, interpretability, and robustness by explicitly modeling the underlying causal structure of data. In the human brain's cognitive process [39], [40], causal relationships are inherently learned, allowing for reliable reasoning and decision-making even under complex conditions [41]. Inspired by this biological mechanism, causal-aware learning [42], [43] seeks to emulate similar behavior by incorporating prior knowledge and reasoning capabilities into neural models. It is worth noting that causal inference refers to a concrete methodological framework for estimating causal effects, while causal-aware modeling represents a broader strategy that guides network design using causal reasoning principles.

A variety of causal-aware approaches have been proposed. Learning [44] introduced a decomposable SCM to estimate causal effects via combinatorial generalization. Zhang et al. [45] designed a causal constraint module using random Fourier features to capture nonlinear dependencies between image regions and labels. Kancheti et al. [46] applied causal regularization by aligning learning outcomes with domain-specific prior functions. Ding et al. [47] further applied causal inference frameworks in reinforcement learning to handle structural uncertainty and unobserved confounders. While these approaches have made significant progress, they still face key limitations. Many rely on static or hand-crafted causal priors that lack adaptability in dynamic, multiscale environments. Causality is often applied at only one level without jointly modeling attention and feature representations, limiting integration depth. Moreover, few methods dynamically refine causal features during training, leading to mismatches with evolving visual contexts. To overcome these issues, our proposed DCDT framework introduces a dual causal-aware

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detection strategy that integrates causality at both the attention-level and the feature-level. Furthermore, the causal prototypes are iteratively refined during training using a contrastive learning strategy, enabling the model to progressively approximate true causal features. These innovations distinguish our method from existing approaches and provide a more flexible and effective framework for robust object detection in complex remote sensing scenarios.

III. Proposed Method

A. Overview of DCDT

The authors propose a novel DCDT model to address the performance bottleneck caused by complex background interference and feature confusion in remote sensing image target detection. DCDT constructs a hierarchical causal-aware feature learning framework by integrating causal inference with local feature enhancement mechanisms. The core innovations of the proposed approach are reflected in the following three aspects.

1) *Dual Causal-Aware Strategy:* A dual causal-aware strategy of attention-level and feature-level is proposed. A multiscale causal prototype bank is introduced into the attention mechanism to guide the model's focus toward causal features. In this process, a causal prior matrix is constructed to quantify the causality between image features and task-essential attributes, which again enhances the causal features.
2) *Local-Global Synergy:* While maintaining global context-awareness, a kernel domain decoupling mechanism is innovatively introduced to enhance the structure-awareness capability of local regions.
3) *Multiscale Optimization:* An efficient feature pyramid is constructed through the integration of window partitioning and progressive patch merging operations. During this process, the causal prototype bank is dynamically updated to enable multiscale feature refinement.

The input RGB images are resized to a uniform resolution of 800×800, standardizing the input size for the model. The patch splitting module then divides the image into nonoverlapping patches, each measuring 4×4. Each patch is treated as a distinct "token," with its feature dimension calculated as $4\times4\times3 = 48$, where the 3 represents the three color channels. These tokens are subsequently flattened into a feature matrix with a dimension of 48. Following the patch partitioning process, the resulting input tensor has dimensions $(H/4 \times W/4 \times 48)$. To adapt these features for further processing, a linear embedding layer is applied, mapping the original feature dimensions to a specified dimension C. The transformed input features, now with dimensions $(H/4 \times W/4 \times C)$, are fed into a hierarchical architecture comprising four stages of feature learning.

As illustrated in Fig. 2, the DCDT model consists of four stages, each comprising a DCDT block followed by a patch merging layer for downsampling. The number of DCDT blocks stacked in each stage is {2, 2, 6, 2}. Each DCDT block integrates two key components: the CPE module and the LKFE module. The functions and detailed architectures of these critical modules are thoroughly discussed in Section III-B and III-C.

Before being processed by the DCDT block, the image features with dimensions $(H/4 \times W/4 \times C)$ are partitioned into N nonoverlapping windows, each of size $w \times w$ (where w is typically set to 25). This window-based segmentation strategy significantly reduces the computational complexity in applying self-attention mechanisms to 2-D image data. We use a larger window division size. This can include more adjacent contextual information in the window and help the model to recognize the object. At the end of the first three stages, a patch merging layer is applied to downsample the image. After three iterations of patch merging, the features within the windows undergo information interaction at a global scale.

B. Causal Prior Extraction

The core idea of the CPE module is to adopt a decomposable causal perspective to analyze the causal relationships between images and target predictions, and to model their causal nature to achieve accurate identification and enhancement of causal features.

1) Causal Relations in SCMs: We construct SCMs as illustrated in Fig. 1, where solid lines represent causal relationships and dashed lines represent potential containment relationships. P denotes the causal constraints imposed on scenes and objects through a library of prototypical features. These constraints facilitate the separation of causal features (Z_C), which are beneficial for outcome prediction, from the overall feature distribution. In remote sensing target detection tasks, image features intuitively contain foreground information (O) and background information (S), each of which includes biases that are either beneficial for target detection (i.e., causal features, C_S and C_O) or detrimental (i.e., noncausal features, N_S and N_O). As shown in Fig. 1(a), these four types of features collectively influence the detection outcome (Y). The goal is to isolate the influence of noncausal features on the detection results. However, the identification of noncausal features in this process is incomplete and cannot isolate all confounding factors in the image, so we focus on causal effects that play a direct role in predicting the outcome. We merge causal features and decompose the overall composite effect into the sum of confounding effects (N_S, N_O) and causal effects (Z_C). The causal features Z_C are extracted by applying causal constraints to the input image X through the causal prototype features P. This process is depicted in Fig. 1(b) and (c). Our causal pathway analysis reveals that scene-object separation is unnecessary for causal effect extraction, as direct image intervention provides a more efficient alternative. We therefore adopt the causal pathway shown in Fig. 1(d) for integrated causal effect computation.

2) Attention-Level Causal Constraint (ACC): We construct corresponding causal prototype features for each scale to eliminate visual bias across multiple scales. The causal prototypes are globally shared within each scale, enabling the model to extract relevant information across multiple windows. The multiscale causal constraint method is illustrated in Fig. 2. As the attention-level causal enhancement, the proposed method introduces explicit causal constraints and a self-updating

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mechanism through the construction of causal feature prototypes, progressively enhancing causally aware representations from the image features. We first construct a multiscale causal prototype bank, which serves as a globally learnable structure comprising four elements, each corresponding to one of the four hierarchical stages in the model. In the initial training phase, 5% of the image samples are selected to initialize a prototype feature library. Multiscale feature representations are extracted using shallow convolutions and adaptive average pooling. These features are subsequently aggregated to form the initial prototype representations at each scale, thereby establishing a hierarchically semantic prototype bank. Each prototype element has a size of (N_i, D_i) and is represented as follows:

$$P_i \in \mathbb{R}^{N_i \times D_i}, \quad i = 1, 2, 3, 4 \tag{1}$$

where P_i denotes the prototype element of the ith stage, N_i represents the number of windows partitioned at that stage, and D_i refers to the dimensionality of the feature vector for each window.

We adopt the causal enhanced window multihead self-attention (Causal W-MSA) mechanism in place of the standard window-based multihead self-attention (W-MSA). In the Causal W-MSA structure, the expressive capacity of the attention mechanism is improved by concatenating the prototype element P with the key K and value V to form an augmented set of key–value pairs. Its attention structure is shown on the right side of Fig. 2. The integration of the prototype feature into the keys and values enables the explicit introduction of causal relationships, allowing the computed attention weights to better reflect the causal dependencies between features. The resulting attention matrix is normalized using a softmax function to ensure a valid probability distribution, and the final output attention is computed as follows:

$$\textbf{Attention} = \text{softmax}\left(\frac{Q \cdot [K, P]^T}{\sqrt{d}} + \mathbf{B}\right) \cdot [V, P] \tag{2}$$

where Q, K, and V denote the query, key, and value vectors, respectively, B is the positional bias matrix, and d is the scaling factor. Through this method, causal information is effectively encoded into the attention matrix, thereby enhancing the model's sensitivity to causal features.

3) Feature-Level Causal Constraint (FCC): The CPE module incorporates causal prototypes into the self-attention mechanism to achieve causal modeling at the attention level, while also playing a pivotal role in establishing FCC. During training, the attention matrix computed in CPE is explicitly decomposed into two components: one representing the self-attention structure of the image, and the other capturing cross-attention between image features and causal prototypes. This cross-attention map serves as a causal prior, quantifying the relevance between spatial regions and task-specific semantics. Regions with stronger causal relevance to the prediction task are assigned higher weights, thereby directing the model's focus toward causally meaningful areas. Subsequently, the causal prior is fused with the image features via element-wise multiplication to generate causally enriched representations, which serve as a form of FCC. This fusion not only strengthens the structural dependency between causal features and prediction outcomes but also suppresses the influence of noncausal factors, thereby enhancing the model's robustness in complex visual environments.

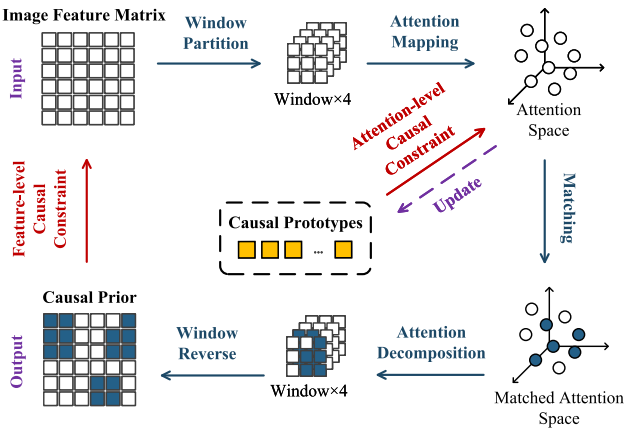


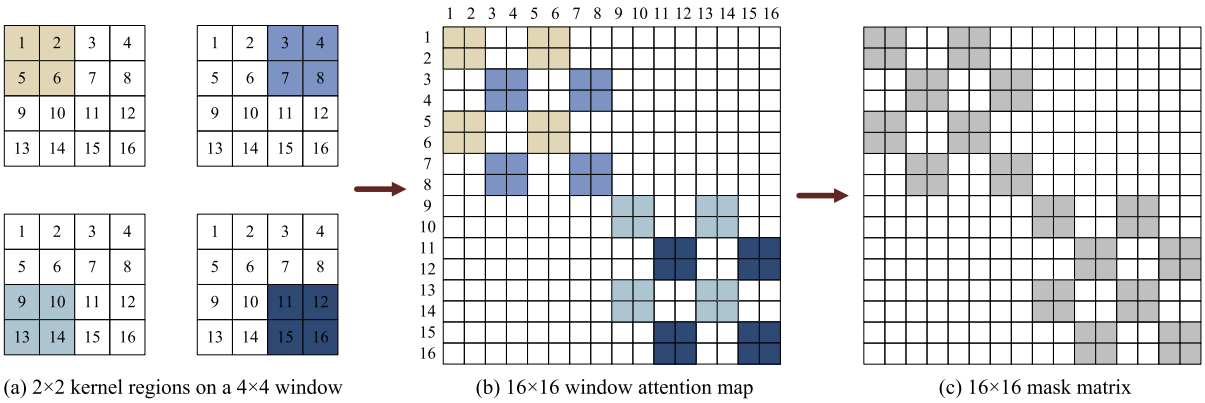
Fig. 3. Dual causal constraints in the CPE module.

The dual-level causal modeling process of the CPE module is illustrated in Fig. 3. It first injects causal information into the attention mechanism for matching. The resulting attention matrix is then decomposed to extract a causal prior. This causal prior is subsequently fused with the feature space to reinforce causally relevant representations, thereby enabling causal perception at both the attention and feature levels. The resulting fused representation, enriched with causal semantics, is further propagated to the LKFE module, where it undergoes additional enhancement through localized structural modeling.

C. Local Kernel-Region Feature Enhancement

1) Definition of Local Kernel Region: The LKFE module is designed to extract and enhance local features within segmented windows. To achieve this, we introduce the kernel-region-based window multihead self-attention mechanism (Kernel W-MSA) as a replacement for the standard W-MSA. The motivation behind the LKFE strategy lies in the strong spatial locality inherent in visual data, where semantically coherent structures typically manifest within compact regions. By dividing each $W \times W$ attention window into $(W/k) \times (W/k)$ nonoverlapping kernel regions and applying a sparse masking scheme, we explicitly incorporate a localization inductive bias that emphasizes causally relevant subregions while suppressing irrelevant background noise. For example, with $W = 25$ and $k = 5$, each attention window is partitioned into a grid of 5×5 local subregions. The kernel size k is empirically chosen to balance spatial granularity with computational efficiency. This nonoverlapping partitioning strategy avoids redundant computation and enables the model to capture independent local structures, while still maintaining access to global context across the entire attention window. The choice of k is task-dependent, and the effects of different kernel sizes are systematically evaluated through ablation studies presented in Section IV-E.

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(a) 2×2 kernel regions on a 4×4 window (b) 16×16 window attention map (c) 16×16 mask matrix

Fig. 4. Mapping of nonoverlapping kernel regions in the global attention matrix. (a) 4×4 window is divided into four nonoverlapping 2×2 kernel regions. (b) Positional mapping of each kernel region within the local self-attention map of the current window. (c) Corresponding SpaM matrix formed by the positional mappings of all kernel regions.

As illustrated in Fig. 4, the spatial positions of kernel regions are mapped into the full attention space, forming a diagonally distributed SpaM matrix. For illustrative purposes, the figure uses a 4×4 window and a kernel size of 2×2 as an example. Each activated region in the mask corresponds to a semantically meaningful receptive field in the image, enabling targeted enhancement while preserving spatial consistency. From a theoretical perspective, the LKFE mechanism bridges the gap between self-attention and convolutional inductive biases. It instantiates localized receptive fields within the transformer's attention graph, analogous to graph-based neighborhood aggregation in graph signal processing.

2) SpaM Matrix: Based on the local kernel-region definition, we incorporate a SpaM computation into W-MSA, constructing Kernel W-MSA. This method generates corresponding SpaMs for different window and kernel sizes. Specifically, the attention values corresponding to kernel region positions are set to 1, while nonkernel positions are set to 0, resulting in a SpaM matrix with the same dimensions as the attention map. The computation is formalized as follows:

$$M_{i,j} = \begin{cases} 1, & \text{if } (i,j) \in \mathcal{R}_k \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where (i, j) denotes the 2-D index in the attention matrix, M_{ij} represents the value at position (i, j) in the mask matrix, and R_k is the set of position indices corresponding to all kernel regions mapped within the attention matrix. This SpaM assigns additional weights to the locations corresponding to kernel regions in the global attention matrix, thereby enhancing the model's focus on local regions. Through the Hadamard product operation between the SpaM and the attention matrix, a nonparametric weight routing mechanism is established, where 1 in the mask guides the attention weights to concentrate within the kernel regions, while 0 serves to shield against attention interference from unrelated areas.

In other words, although attention is computed over the entire window, the mask explicitly activates and amplifies local attention connections, which are subsequently fused with the original attention map. This design maintains the global contextual modeling ability while reinforcing the expressiveness of local structures, effectively embedding a locality prior into the parameter space. Consequently, the model acquires enhanced "neighbor sensitivity" within the self-attention mechanism, facilitating the capture of spatially continuous local structures in images. The computation of the enhanced attention is given by

$$\begin{aligned} A_{\text{orig}} &= \text{Softmax}\left(\frac{Q \cdot K^T}{\sqrt{d}}\right) \\ A_{\text{masked}} &= \text{Softmax}\left(\frac{Q \cdot K^T \odot M}{\sqrt{d}}\right) \\ A_{\text{fused}} &= \lambda \cdot A_{\text{masked}} + (1-\lambda) \cdot A_{\text{orig}} \end{aligned} \tag{4}$$

where M is the SpaM matrix, Q and K denote the query and key vectors, respectively, and $\sqrt{d}$ is the scaling factor. $\odot$ representing Hadamard product calculation. A_{orig} denotes the conventional self-attention matrix computed via the dot-product between Q and K, which is subsequently spatially constrained by M to yield the locally enhanced attention matrix A_{masked}. $\lambda \in [0, 1]$ is a fusion coefficient, set to 0.5 by default. The resulting fused matrix A_{fused} represents the final output of this operation.

3) Module Output Fusion: The features output by the CPE and LKFE modules are fused, and the fusion process is defined as follows:

$$H = \alpha \cdot H_{\text{CPE}} + (1-\alpha) \cdot H_{\text{LKFE}} \tag{5}$$

where H_{CPE} denotes the causally enhanced features from the CPE module, H_{LKFE} represents the locally enhanced features from the LKFE module, and $\alpha \in [0, 1]$ is the fusion coefficient, set to 0.5 by default. This design enables the model to maintain a strong focus on critical causal regions while effectively integrating local contextual dependencies, thereby achieving collaborative modeling of global and local causal features. As a result, the model's ability to accurately localize and recognize targets in complex remote sensing scenes is significantly enhanced.

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D. Patch Merging

Patch merging is adopted to achieve multiscale feature extraction and spatial dimension compression. Serving as a downsampling module, this operation is embedded between two consecutive stages, aggregating adjacent 2×2 patches to reduce the resolution of the feature map while increasing the channel dimension. Specifically, given an input feature map $X \in \mathbb{R}^{H \times W \times C}$, adjacent 2×2 patches are concatenated along the channel dimension to form an intermediate feature map of size $\mathbb{R}^{(H/2) \times (W/2) \times 4C}$, followed by a linear projection that reduces the channel dimension to $2C$. The proposed model adopts a four-stage hierarchical structure. The input image of size $H \times W$ is first partitioned into nonoverlapping 4×4 patches and embedded via a linear projection, resulting in an initial token map of size $(H/4) \times (W/4) \times C$. Subsequently, three patch merging operations are applied to progressively halve the spatial dimensions, with the output feature sizes after each merging operation being $(H/8) \times (W/8) \times 2C$, $(H/16) \times (W/16) \times 4C$, and $(H/32) \times (W/32) \times 8C$, respectively.

It can thus be seen that, by the final stage, the feature map size has been reduced to $(H/32) \times (W/32) \times 8C$. At this point, the input tokens have undergone successive patch merging operations and encode multiscale contextual information, thereby acquiring the capability for global semantic representation. In our experiments, the input images are resized uniformly to 800×800. Consequently, the input feature map at the final stage has dimensions of 25×25. At this stage, the window size exactly matches the feature map size, enabling the window to fully cover the entire feature map. This design ensures that, by the final layer, the attention mechanism can perform global modeling across all tokens in the feature space, effectively achieving comprehensive global information integration.

E. Model Optimization

The multiscale causal prototype bank interacts with image features at each stage to enhance their attention and enforce a causal representation. To align the causal prior (cp) with the ground-truth bounding box labels, we introduce a causal loss (L_{causal}). This loss quantifies the discrepancy between the causal prior matrix and the corresponding ground-truth mask matrix. Specifically, the causal prior represents the attention distribution, while the bounding box mask is generated based on ground-truth labels, where pixel points inside the bounding box are labeled as 1, and those outside are labeled as 0. The resulting causal loss function is defined as follows:

$$\mathcal{L}_{\text{causal}} = \frac{1}{N} \sum_{i=1}^{N} \left(\text{cp}_i - \text{mask}_i\right)^2 \tag{6}$$

where L_{causal} represents the causal constraint loss, cp is the causal prior matrix computed between the image features and the prototype bank within the attention mechanism, and $\text{cp}_i \in [0, 1]$ denotes the ith element in the causal prior matrix. mask is the $0-1$ matrix generated from ground-truth bounding box labels, with $\text{mask}_i \in \{0, 1\}$ indicating whether a spatial location falls inside a ground-truth region. $N = n \times n$ denotes the total number of spatial positions.

To progressively refine the prototype representations, we further introduce a contrastive learning optimization strategy. Specifically, during training, image regions corresponding to ground-truth boxes are treated as positive samples, while nonobject regions are treated as negative samples. The causal prior acts as a soft attention mask, guiding prototype features to minimize the distance to positive features and maximize the separation from negatives. This contrastive scheme operates implicitly via gradient supervision from the causal loss L_{causal}, encouraging each prototype to evolve toward a distinct and discriminative representation of causal object features. As a result, the prototypes are iteratively updated to encode more compact, object-specific semantics throughout training.

The overall loss consists of region proposal network (RPN) and region of interest (ROI) losses, according to the structure of faster R-CNN [48]. Specifically, the RPN loss (L_{RPN}) includes a classification loss (cross-entropy) and a regression loss (SmoothL1) to optimize proposal quality, while the ROI loss (L_{ROI}) combines a bounding box classification loss and a regression loss, similarly calculated by cross-entropy and SmoothL1, respectively. The final objective loss function is the sum of the losses from the RPN and the ROI Head, and the loss arising from causal optimization

$$L = \lambda_c \cdot L_{\text{causal}} + L_{\text{RPN}} + L_{\text{ROI}} \tag{7}$$

here, λ_c is a tunable hyperparameter that controls the contribution of the causal prior alignment to the overall optimization. Its value is empirically set to 0.1.

This overall loss is jointly optimized to improve object detection performance while providing a more causal feature representation with a prototype bank.

IV. Experiments

To validate the effectiveness and robustness of our proposed DCDT, we conducted comprehensive experiments on two large-scale public remote sensing datasets: DIOR and HRRSD. To evaluate the DCDT's performance in the task of object detection from remote sensing imagery, we compared our method with both classical and state-of-the-art detection approaches. The comparative methods include several classical CNN-based detectors, such as the two-stage detection models Faster RCNN [48], Mask RCNN [49], Cascade RCNN [50], and DConvTransLGA [51]. These models are known for their high detection accuracy and robust performance in general object detection tasks. Additionally, one-stage CNN-based detectors such as SSD [52], RetinaNet [53], YOLOv3 [54], YOLOv5, M2Det [55], EfficientDet [56], and YOLOv4+MFDF* [57] were evaluated. These models prioritize faster detection via an end-to-end detection architecture. In addition, transformer-based methods such as DETR [58], Anchor DETR [59], Swin Transformer [60], QETR [61], and GOCPK [62] for remote sensing detection are also included in the comparison. These models leverage the transformer framework to capture global contextual features and achieve more accurate detection results. In addition, we constructed a baseline model with an architecture similar to DCDT, but using only the standard W-MSA mechanism without introducing any causal-based improvements or local enhancements. It

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can be used as a reference baseline to evaluate the model's performance.

A. Datasets

We evaluate our model on two widely used remote sensing datasets: DIOR [63][1] and TGRS-HRRSD [64][2], both with standardized image resolutions.

1) DIOR: DIOR is a large-scale optical remote sensing dataset containing 23 463 images and 192 472 labeled instances across 20 categories such as airplanes and baseball fields. A significant proportion of objects are small or medium in size, making DIOR particularly suitable for evaluating performance on small-scale object detection.

2) TGRS-HRRSD: TGRS-HRRSD consists of 21 761 images and 57 137 instances across 13 categories, including ships and bridges. Compared to DIOR, it features a higher proportion of medium and large objects, providing a complementary benchmark to assess performance on larger scale targets in complex scenes.

For all experiments, we reported performance using standardized mean accuracy (mAP) as an evaluation metric.

B. Causal Prior Analysis

In the CPE module, the model computes causal prior features for each stage, which are progressively refined through a multiscale prototype bank. The prototype features are iteratively updated by calculating the loss of the causal prior matrix relative to the ground truth, ensuring that each prototype aligns more closely with task-specific causal features in the image. The final visualization of the causal priors is shown in Fig. 5. As observed, in the initial two stages, the learning of high-resolution images enables the causal prototype features to capture primarily edge textures, details, and low-level visual features. These features help the model identify essential information such as edges and shapes. In the deeper stages, as the resolution progressively decreases, the model shifts its focus to high-level features. The causal prototypes increasingly emphasize global structure, semantic information, and the overall representation of the object, with reduced interference from background noise. These high-level features are crucial for task decision-making, as they facilitate the model's understanding of object relationships and the global scene structure, thereby enhancing prediction accuracy. The causal prior serves as an interpretable guidance signal during feature selection and enhancement. By modulating image features at both the attention and representation levels, it enforces consistency between the model's attention allocation and the true object semantics, leading to more robust and generalizable representations.

C. Implementation Details

Our experimental environment consists of the PyTorch framework on the Ubuntu 20.04 operating system. All experiments were conducted using two NVIDIA GeForce RTX 3090 GPUs, each with 24 GB of memory. The network utilizes Swin-T pretrained on the large-scale ImageNet-1K dataset as the initialization model. The input images were uniformly resized to 800×800 pixels, and no additional data augmentation techniques were employed. The AdamW optimizer was used for training, with a learning rate of 6×10^{-5}, momentum parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and a weight decay of 0.05. A two-stage learning rate scheduler was implemented during training: The first stage employed a linear learning rate adjustment strategy from step 0 to step 1000, with an initial factor of 0.01. The second stage applied a cosine annealing strategy from epoch 0 to epoch 30, with the final learning rate reduced to 3×10^{-6}. To ensure comprehensive training, a 30-epoch training schedule was adopted with a batch size of 2. To enhance the model's feature representation capabilities, the drop path rate was set to 0.6.

[1]https://ieee-dataport.org/documents/dior

[2]https://github.com/CrazyStoneonRoad/TGRS-HRRSD-Dataset

TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DIOR DATASET. THE BOLDED DATA ARE THE BEST RESULTS

Method	mAP	$AP_{50:95}$	AP_{75}	AP_s	AP_m	AP_l
Faster RCNN [48]	56.9	32.6	34.0	6.6	26.9	49.3
Mask RCNN [49]	57.0	33.1	33.9	6.8	27.5	49.3
Cascade RCNN [50]	57.5	35.4	37.4	6.7	29.1	53.9
RetinaNet [53]	55.8	35.4	37.4	5.9	28.2	53.3
SSD [52]	55.0	31.1	32.0	1.6	23.9	49.8
YOLOv3 [54]	59.3	37.2	39.5	9.0	31.8	53.7
YOLOv5	61.5	35.8	37.7	7.2	27.8	52.9
M2Det [55]	57.6	-	-	-	-	-
DETR [58]	58.2	38.0	40.4	3.6	27.3	61.8
EfficientDet D2 [56]	61.5	-	-	-	-	-
Swin [60]	64.7	40.0	42.6	8.5	31.7	58.3
AnchorDETR [59]	56.4	31.6	31.6	2.8	23.9	50.1
YOLOv4+MFDF* [57]	63.9	33.8	-	8.6	30.0	47.4
DConvTransLGA [51]	61.3	36.3	-	-	-	-
Baseline	58.8	33.3	33.6	7.1	26.6	47.7
DCDT	**65.6**	**45.1**	**49.7**	**9.3**	**35.1**	**65.4**

D. Experimental Results and Analysis

We evaluated our method with other classical object detection methods on DIOR and HRRSD datasets, and the experimental results are presented in Tables I and II.

1) Results on DIOR Dataset: According to the experimental results presented in Table I, the proposed DCDT network achieves a remarkable performance on the DIOR dataset, attaining a mAP of 65.6%. Compared to CNN-based models, DCDT outperforms Faster R-CNN by 8.7%, Cascade R-CNN by 8.1%, and YOLOv4+MFDF* by 1.7%. It also yields improvements of 8.0% and 4.1% over single-stage CNN detectors such as M2Det and EfficientDet-D2, respectively. Furthermore, DCDT surpasses the performance of the deformable convolutional transformer DConvTransLGA by 4.3%. These results demonstrate the strong feature representation capability of DCDT and validate its effectiveness in remote sensing object detection tasks.

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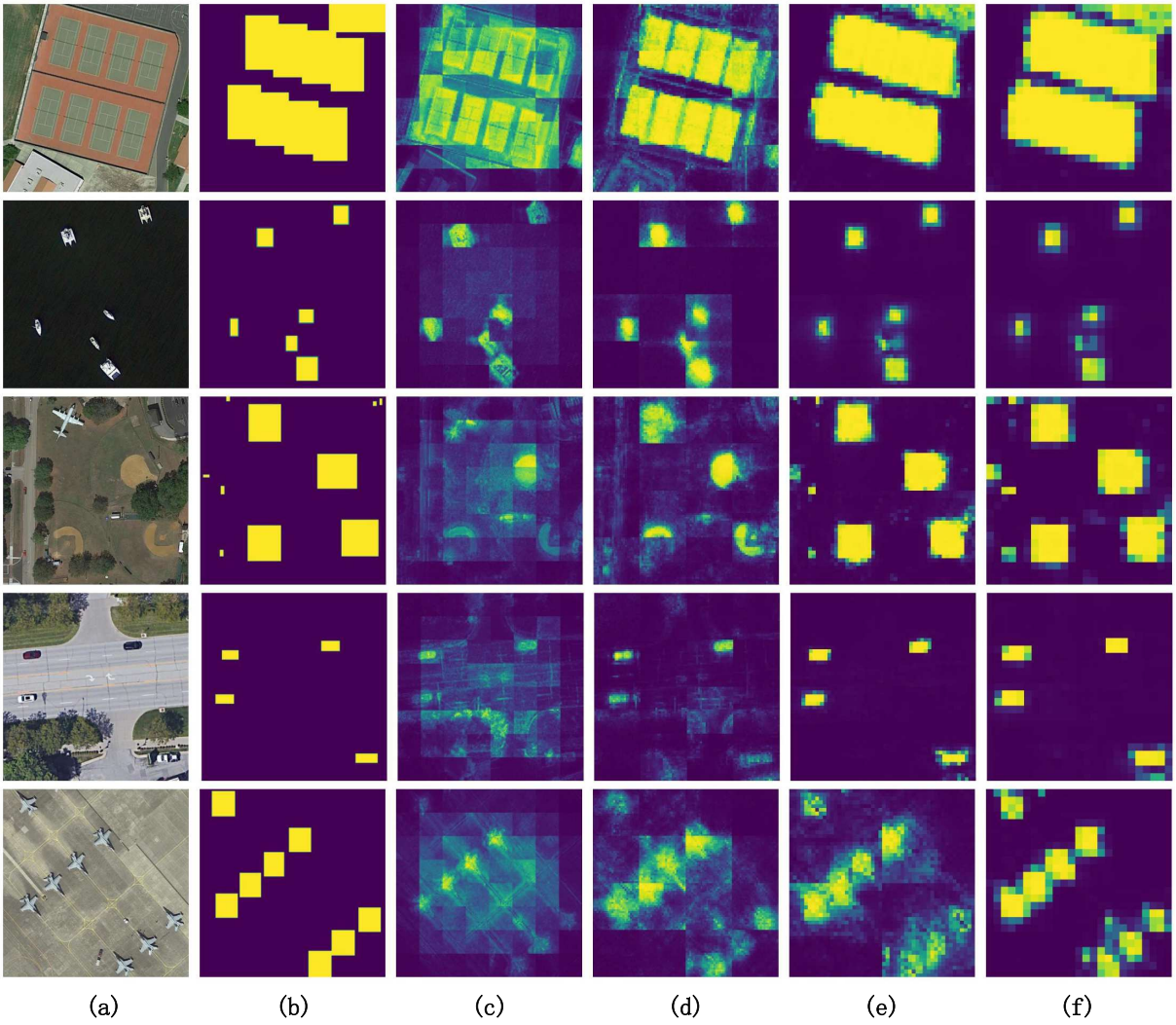


Fig. 5. Visualization of causal priors. (a) Original input image. (b) Bounding box label visualization. (c)–(f) Visualizations of causal priors corresponding to the four stages of the model.

TABLE II

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE HRRSD DATASET. THE BOLDED DATA ARE THE BEST RESULTS

Method	mAP	$AP_{50:95}$	AP_{75}	AP_s	AP_m	AP_l	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12	H13
Faster RCNN [48]	84.0	47.8	48.5	12.8	41.4	45.7	86.1	82.6	96.3	93.2	61.2	90.1	97.5	84.7	90.1	90.4	92.5	70.7	55.9
Mask RCNN [49]	84.7	50.2	54.0	15.0	45.4	48.5	87.6	84.2	97.2	93.4	60.7	89.9	96.8	88.3	91.8	91.6	92.0	67.8	59.5
Cascade RCNN [50]	84.6	53.7	59.7	8.4	46.7	51.8	89.6	84.4	96.4	94.9	61.9	91.2	96.9	84.3	91.0	92.7	90.4	71.2	55.4
RetinaNet [53]	83.4	50.7	55.0	12.3	43.7	49.5	87.2	86.3	96.8	93.2	55.7	86.2	97.4	89.6	88.9	86.4	90.5	73.3	52.1
SSD [52]	79.8	45.4	46.5	3.7	37.1	45.7	81.0	87.7	96.8	93.1	43.8	81.8	97.3	84.9	87.0	71.5	91.9	76.8	44.1
YOLOv3 [54]	86.1	56.1	63.1	21.8	48.7	54.6	84.7	84.2	97.0	93.7	72.3	93.6	96.8	87.3	92.6	90.0	90.6	76.4	60.7
YOLOv5	88.4	60.4	69.2	14.4	50.7	58.6	91.1	87.7	97.5	94.7	68.8	93.0	97.9	89.6	93.3	93.5	93.2	81.2	67.3
DETR [58]	88.0	59.9	68.5	22.8	47.9	59.5	93.0	90.5	97.1	94.2	68.0	92.9	98.6	92.0	92.3	91.6	85.6	75.0	72.5
Swin [60]	90.9	59.5	67.6	23.1	52.3	57.5	93.3	93.4	98.5	95.8	72.0	93.6	**98.8**	**93.3**	**96.2**	95.6	94.8	85.8	71.0
AnchorDETR [59]	87.6	53.6	58.7	22.6	44.5	54.1	92.0	90.7	98.0	91.7	64.3	89.6	97.6	91.5	92.7	88.4	93.6	80.2	67.7
QETR [61]	91.1	-	-	-	-	-	**98.7**	**95.9**	69.6	**97.6**	**97.7**	**99.0**	97.8	78.7	92.6	91.3	91.2	**88.5**	**85.7**
GOCPK [62]	91.0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
DConvTransLGA [51]	82.1	49.2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Baseline	86.6	52.7	57.3	15.5	46.5	50.5	88.8	90.5	98.1	94.7	58.7	93.0	98.3	90.8	90.7	92.3	89.4	76.1	63.9
DCDT	**91.4**	**64.4**	**74.6**	**30.2**	**56.0**	**62.3**	93.9	92.8	**98.7**	95.7	74.6	94.3	98.0	93.0	94.8	**96.8**	**95.6**	86.4	73.1

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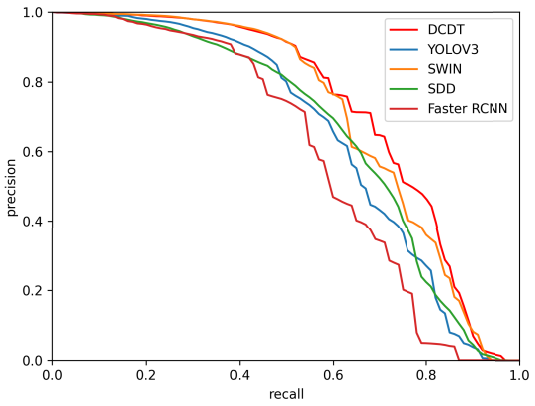


Fig. 6. P-R curves for different models on the DIOR dataset.

Compared with transformer-based architectures, DCDT exhibits substantial gains, improving mAP by 7.4% over DETR, 9.2% over Anchor DETR, 0.9% over Swin Transformer, and 6.8% over the baseline transformer model. These improvements underscore the efficacy of the causal prior-driven design in DCDT, which enhances the model's ability to capture essential spatial dependencies and boosts overall detection accuracy.

In terms of scale-wise detection performance, DCDT achieves notable improvements over the baseline, with mAP increases of 17.7%, 8.5%, and 2.2% for large, medium, and small objects, respectively. These results highlight the benefits of integrating multiscale causal priors and localized fine-grained feature enhancement. The model not only improves the discriminability of object boundaries in spatially constrained regions but also enhances its sensitivity to objects across varying scales. DCDT also delivers significant gains under stricter evaluation metrics, with increases of 11.8% in $AP_{50:95}$ and 16.1% in AP_{75} over the baseline model—both exceeding the improvement observed in mAP. Since higher intersection-over-union (IoU) thresholds require more precise alignment between predicted and ground-truth boxes, these results indicate that DCDT, through the integration of causal priors, provides informative constraints that guide attention toward task-relevant regions, thereby substantially improving bounding box localization accuracy.

To further validate the model's performance across varying object scales, precision-recall (P-R) curves were plotted for several representative models, as illustrated in Fig. 6. The P-R curve illustrates the trade-off between precision and recall, where a curve that remains close to the top-right corner indicates both high precision and high recall. The plotted curves show that DCDT consistently maintains high precision across all levels of recall. This behavior indicates that the model effectively suppresses false positives while preserving high recall, demonstrating its robustness and suitability for complex remote sensing detection scenarios.

2) Results on HRRSD Dataset: The instance size distribution of the HRRSD dataset is similar to that of natural datasets, so we chose it as one of the datasets for evaluating the model results. As shown in Table II, the proposed DCDT model achieves a mAP of 91.4%, outperforming several CNN-based detectors. Specifically, it surpasses Faster R-CNN and Cascade R-CNN by 7.4% and 6.8%, respectively, and improves upon DConvTransLGA by 9.3%. Compared to one-stage CNN-based detectors, DCDT delivers significant gains of 8.0% over RetinaNet and 5.3% over YOLOv3. When compared with transformer-based detectors, DCDT also demonstrates consistent improvements: it outperforms the baseline transformer model by 4.8%, DETR by 3.4%, Anchor DETR by 3.8%, GOCPK by 0.4%, Swin Transformer by 0.5%, and QETR by 0.3%. These performance gains highlight the effectiveness of the proposed sparse attention mechanism in focusing on object-relevant regions, as well as the contribution of the causal prior module in enhancing the modeling of object-context relationships.

In addition, DCDT achieves significant improvements in high-precision metrics: the $AP_{50:95}$ and AP_{75} scores exceed those of the baseline by 11.7% and 17.3%, respectively. These enhancements, which are more substantial than those observed in mAP, suggest that DCDT is particularly effective at achieving accurate object localization under stricter confidence thresholds. In terms of scale-wise detection performance, DCDT achieves notable mAP improvements of 11.8%, 9.5%, and 14.7% over the baseline for large, medium, and small objects, respectively. The significant gains on both large- and small-scale targets indicate that the proposed local enhancement module effectively captures critical regional details, while the model's global modeling capability remains robust and highly effective.

Overall, the consistent performance of DCDT across both the DIOR and HRRSD datasets demonstrates its strong adaptability and robustness. On the DIOR dataset, which features a high proportion of small objects, DCDT significantly enhances the model's capacity to detect fine-grained targets. On the HRRSD dataset, the synergy between the sparse attention mechanism and the causal prior further improves detection accuracy. These results confirm that DCDT is a high-performance and generalizable model, well-suited for diverse remote sensing object detection applications.

E. Ablation Study

We performed a systematic ablation study on the DIOR dataset to evaluate the efficacy of the three proposed key components: ACC, FCC, and LKFE. These modules are designed to jointly improve the causal modeling and local feature representation capabilities of our framework. It is important to note that FCC is dependent on ACC, as a feature-level causal constraint (FCC) requires a causal prior derived from attention-level interactions. Therefore, FCC cannot be evaluated independently without ACC. To evaluate the individual and combined contributions of the proposed modules, we design six experimental variants based on different combinations of ACC, FCC, and LKFE.

1) *Baseline:* This variant serves as the reference model. It employs a plain transformer backbone utilizing the standard W-MSA mechanism, without any causal or locality-enhancing modifications.

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TABLE III
ABLATION STUDY EVALUATING THE INDIVIDUAL AND COMBINED CONTRIBUTIONS OF ACC, FCC, AND LKFE ON DETECTION PERFORMANCE

Module	ACC	FCC	LKFE	mAP	$AP_{50:95}$	AP_{75}	AP_s	AP_m	AP_l
BASE				58.8	33.3	33.6	7.1	26.6	47.7
ACC-only	✓			63.2	41.7	46.1	8.3	32.6	61.1
LKFE-only			✓	62.6	38.8	41.5	7.6	28.4	57.3
ACC & LKFE	✓		✓	64.2	43.0	47.2	8.6	32.5	62.7
ACC & FCC	✓	✓		63.9	42.9	46.9	8.3	32.7	62.5
ACC & FCC & LKFE	✓	✓	✓	**65.6**	**45.1**	**49.7**	**9.3**	**35.1**	**65.4**

2) *ACC-only:* In this setting, the standard W-MSA is replaced with Causal W-MSA, which incorporates ACC by integrating the multiscale causal prototype bank into the attention mechanism.
3) *LKFE-only:* This configuration activates only the LKFE module. Here, the standard W-MSA is replaced by Kernel W-MSA, which enhances local feature modeling through kernel-region partitioning and sparse attention masking.
4) *ACC and LKFE:* This variant combines the benefits of both attention-level causal modeling and localized structural enhancement. Specifically, image features are independently processed by Causal W-MSA and Kernel W-MSA branches, without incorporating FCC. The outputs from both branches are fused using a weighted sum with the fusion coefficient α set to 0.5 by default.
5) *ACC and FCC:* This model variant evaluates the combined effect of ACC and FCC. First, a causal prior is extracted via Causal W-MSA (ACC). This prior is then fused with the original image features to construct causally enriched representations, which are subsequently processed by standard W-MSA. The final output is obtained by fusing the outputs of the Causal W-MSA and W-MSA branches. LKFE is not used in this variant.
6) *ACC and FCC and LKFE:* This is the complete model incorporating all three proposed modules. The attention-level causal prior obtained from ACC is used to modulate the image features through FCC, resulting in feature representations that are both semantically and causally enriched. These features are then input into Kernel W-MSA for localized enhancement (LKFE). Finally, the outputs from both the Causal W-MSA and Kernel W-MSA branches are fused to produce the final representation.

The results of the ablation experiments are summarized in Table III. Introducing ACC alone leads to an improvement in mAP from 58.8% to 63.2%, confirming that integrating causal prototypes into the attention computation enhances semantic relevance and suppresses spurious correlations. The LKFE module independently yields a 62.6% mAP, demonstrating the effectiveness of kernel-based window partitioning and sparse masking in capturing spatially coherent local patterns. When ACC and LKFE are combined, the model achieves 64.2% mAP, confirming their complementary roles. After adding FCC to ACC, the mAP is 63.9%. This indicates that the combination of the two causal constraints can further enhance the model's focus on causal features. The full model, integrating ACC, FCC, and LKFE, achieves the best overall performance of 65.6% mAP, with consistent improvements across all object sizes. This highlights the synergistic effect of global causal guidance, feature-level modulation, and local structure enhancement in improving detection accuracy.

TABLE IV
ABLATION STUDY ON KEY HYPERPARAMETERS OF DCDT

	DCDT	mAP	$AP_{50:95}$	AP_s	AP_m	AP_l
α	0.3	65.2	**45.3**	**9.4**	34.9	**66.1**
	0.5	**65.6**	45.1	9.3	**35.1**	65.4
	0.7	64.9	44.9	8.7	33.3	66.0
k	5	65.1	44.9	**9.4**	34.6	**65.5**
	7	**65.6**	**45.1**	9.3	**35.1**	65.4
	9	64.7	44.5	9.1	34.5	64.9
R	1%	65.3	44.6	8.7	34.2	65.6
	5%	**65.6**	**45.1**	**9.3**	**35.1**	65.4
	10%	65.5	44.7	8.9	34.1	**65.7**
λ_c	0.1	**65.6**	**45.1**	**9.3**	**35.1**	**65.4**
	0.3	64.6	42.0	8.7	32.6	61.3
	0.5	62.4	41.3	7.8	31.9	61.2

To validate the effectiveness and robustness of the proposed DCDT model, we conduct a comprehensive ablation study on four critical hyperparameters: 1) the feature fusion weight α [see (5)] used to combine outputs from the CPE and LKFE modules; 2) the kernel size k in the LKFE module; 3) the sampling rate R employed during the initialization of the causal prototype bank; and 4) the loss weight λ_c [see (7)] associated with the causal prior alignment term in the overall objective function. The detailed results are reported in Table IV. Experimental results demonstrate that setting α to 0.5 and the kernel size k to 7 achieves the best overall detection performance, indicating an optimal balance between global causal and local structural features. For the causal loss weight, λ_c = 0.1 yields the most favorable outcomes, while larger values lead to performance degradation due to over-constrained optimization. Moreover, a sampling rate of 5% during prototype initialization offers the best trade-off between computational cost and prototype diversity, enabling DCDT to maintain stable and superior performance with minimal overhead.

F. Computational Complexity Analysis

The computational efficiency of the proposed DCDT model is comprehensively evaluated through three key metrics: trainable parameters (Params), floating-point operations (FLOPs), and inference speed (FPS), with detailed results presented

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TABLE V
COMPARISON OF MODEL COMPLEXITY IN TERMS OF FLOPS, PARAMETERS AND FPS FOR VARIOUS OBJECT DETECTION FRAMEWORKS

Method	FLOPs(G)	Params(M)	FPS
Faster RCNN	197.63	41.75	10
Cascade RCNN	226.30	69.39	9
RetinaNet	217.90	36.80	21
CenterNet	95.47	19.13	16
SSD	31.37	26.28	32
YOLOv3	70.04	61.62	12
Swin	143.36	44.84	10
Baseline	140.76	44.33	11
DCDT	192.75	45.18	7

in Table V. The parameter count is comparable to the Swin Transformer and lower than many CNN-based models, indicating manageable complexity despite advanced mechanisms. FLOPs show a slight increase compared to the Swin Transformer but remain below most CNN-based and DETR models, demonstrating minimal computational overhead. However, due to the integration of multiple attention and causal prior modules, DCDT operates at 7 FPS, slightly lower than most compared methods. This trade-off highlights the model's emphasis on robust and accurate detection in complex remote sensing scenarios. Overall, the complexity analysis confirms that DCDT achieves a reasonable balance between computational cost and detection performance, validating its architectural design for high-precision object recognition tasks.

V. CONCLUSION

In this work, we present a novel DCDT for robust object detection in remote sensing images by addressing the pervasive challenge of dataset bias through the lens of causal learning. Based on the theory of SCMs, DCDT introduces a dual-layer causal-aware architecture that incorporates both ACC and FCC. This end-to-end causal learning framework effectively enhances the weighting of task-relevant object features while suppressing noncausal interference that is irrelevant to the target task. In addition, we propose a novel LKFE mechanism that strengthens fine-grained regional features without compromising the global contextual representation. Extensive experiments conducted on benchmark remote sensing datasets validate the superior detection performance of DCDT. These findings demonstrate the effectiveness and generalizability of causal modeling in enhancing the robustness of visual recognition systems.

REFERENCES

[1] C. Wang et al., "Limited-data SAR ATR causal method via dual-invariance intervention," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, doi: 10.1109/tgrs.2025.3528464.

[2] Y. Wei, Z. Pan, and Y. Wu, "Class bias correction matters: A class-incremental learning framework for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, doi: 10.1109/tgrs.2025.3553141.

[3] M. Yin, B. Wang, and C. X. Ling, "MABR: Multilayer adversarial bias removal without prior bias knowledge," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 24, pp. 25724–25732.

[4] Z. Wang et al., "Fairness-aware adversarial perturbation towards bias mitigation for deployed deep models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10369–10378.

[5] P. Singhal et al., "Domain adaptation for bias mitigation in affective computing: Use cases for facial emotion recognition and sentiment analysis systems," *Discover Appl. Sci.*, vol. 7, no. 4, p. 229, Mar. 2025.

[6] Q. Zhang et al., "Dense attention fluid network for salient object detection in optical remote sensing images," *IEEE Trans. Image Process.*, vol. 30, pp. 1305–1317, 2021.

[7] Y. Qiu, Y. Sun, J. Mei, and J. Xu, "Deeply hybrid contrastive learning based on semantic pseudo-label for salient object detection in optical remote sensing images," *IEEE Trans. Multimedia*, vol. 26, pp. 10892–10907, 2024.

[8] Y. Zhang, C. Lan, H. Zhang, G. Ma, and H. Li, "Multimodal remote sensing image matching via learning features and attention mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/tgrs.2023.3348980.

[9] P. Chen, Q. Li, Q. Li, and Z. Wu, "Remote sensing image object detection method with feature denoising fusion module," in *Proc. IEEE 7th Adv. Inf. Technol., Electron. Autom. Control Conf. (IAEAC)*, Mar. 2024, pp. 207–211.

[10] J. Liu, S. Li, C. Zhou, X. Cao, Y. Gao, and B. Wang, "SRAF-Net: A scene-relevant anchor-free object detection network in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/tgrs.2021.3124959.

[11] X. Yang, J. Yan, W. Liao, X. Yang, J. Tang, and T. He, "SCRDet++: Detecting small, cluttered and rotated objects via instance-level feature denoising and rotation loss smoothing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2384–2399, Feb. 2023.

[12] K. R. M. Fernando and C. P. Tsokos, "Dynamically weighted balanced loss: Class imbalanced learning and confidence calibration of deep neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 7, pp. 2940–2951, Jul. 2022.

[13] Z. Yu, H. Huang, W. Chen, Y. Su, Y. Liu, and X. Wang, "YOLO-FaceV2: A scale and occlusion aware face detector," *Pattern Recognit.*, vol. 155, Nov. 2024, Art. no. 110714.

[14] R. Yao, K. Li, Y. Dou, Z. Xu, and M. Wang, "GNP-WGAN: Generative nonlocal a priori augmented Wasserstein generative adversarial networks for seismic data reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/tgrs.2024.3431243.

[15] Y. Li et al., "LSKNet: A foundation lightweight backbone for remote sensing," *Int. J. Comput. Vis.*, vol. 133, no. 3, pp. 1–22, Mar. 2025.

[16] H. Li, B. Zou, L. Zhang, and J. Qin, "CausalCD: A causal graph contrastive learning framework for self-supervised SAR image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/tgrs.2024.3427848.

[17] H. Dong et al., "A novel causal inference-guided feature enhancement framework for PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/tgrs.2023.3343380.

[18] X. Tang, J. Zhang, Y. Xia, and H. Xiao, "DBW-YOLO: A high-precision SAR ship detection method for complex environments," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7029–7039, 2024.

[19] Z. Chen, C. Liu, V. Filaretov, and D. Yukhimets, "Multi-scale ship detection algorithm based on YOLOv7 for complex scene SAR images," *Remote Sens.*, vol. 15, no. 8, p. 2071, Apr. 2023.

[20] J. Liu, R. Lin, G. Wu, R. Liu, Z. Luo, and X. Fan, "CoCoNet: Coupled contrastive learning network with multi-level feature ensemble for multi-modality image fusion," *Int. J. Comput. Vis.*, vol. 132, no. 5, pp. 1748–1775, May 2024.

[21] A. Khan, M. Khan, W. Gueaieb, A. E. Saddik, G. De Masi, and F. Karray, "CamoFocus: Enhancing camouflage object detection with split-feature focal modulation and context refinement," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 1423–1432.

[22] R. Liu, H. Liu, G. Li, H. Hou, T. Yu, and T. Yang, "Contextual debiasing for visual recognition with causal mechanisms," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12755–12765.

[23] W. Qin, H. Zhang, R. Hong, E.-P. Lim, and Q. Sun, "Causal interventional training for image recognition," *IEEE Trans. Multimedia*, vol. 25, pp. 1033–1044, 2023.

[24] J. Pearl, "Causal inference in statistics: An overview," Dept. Comput. Sci., Univ. California, Los Angeles, CA, USA, Tech.~Rep. R-350, 2009.

[25] M. S. Hussain, M. J. Zaki, and D. Subramanian, "Global self-attention as a replacement for graph convolution," in *Proc. 28th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2022, pp. 655–665.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 20:23:30 UTC from IEEE Xplore. Restrictions apply.

[26] Y. Ye and S. Ji, "Sparse graph attention networks," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 905–916, Jan. 2023.
[27] Q. Liu, P. Gao, K. Han, N. Liu, and W. Xiang, "Degradation-aware self-attention based transformer for blind image super-resolution," *IEEE Trans. Multimedia*, vol. 26, pp. 7516–7528, 2024.
[28] Y. Yang et al., "Lglformer: Local–global lifting transformer for remote sensing scene parsing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024.
[29] Y. Zeng, Y. Chen, X. Yang, Q. Li, and J. Yan, "ARS-DETR: Aspect ratio-sensitive detection transformer for aerial oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/tgrs.2024.3364713.
[30] Z. Hu et al., "EMO2-DETR: Efficient-matching oriented object detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/tgrs.2023.3300154.
[31] Y. Guo, H. Wu, S. Yang, and Z. Cai, "Crater-DETR: A novel transformer network for Crater detection based on dense supervision and multiscale fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, doi: 10.1109/tgrs.2024.3376398.
[32] B. N. Patro, V. P. Namboodiri, and V. S. Agneeswaran, "SpectFormer: Frequency and attention is what you need in a vision transformer," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Feb. 2025, pp. 9543–9554.
[33] Z. Yuan et al., "DiTFastAttn: Attention compression for diffusion transformer models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 1196–1219.
[34] X. Xu et al., "An improved Swin transformer-based model for remote sensing object detection and instance segmentation," *Remote Sens.*, vol. 13, no. 23, p. 4779, Nov. 2021.
[35] Z. Xu, D. Wu, C. Yu, X. Chu, N. Sang, and C. Gao, "SCTNet: Single-branch CNN with transformer semantic information for real-time segmentation," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 6, pp. 6378–6386.
[36] Z. Li et al., "ScribFormer: Transformer makes CNN work better for scribble-based medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 43, no. 6, pp. 2254–2265, Jun. 2024.
[37] W. Zhang, Z. Tan, Q. Lv, J. Li, B. Zhu, and Y. Liu, "An efficient hybrid CNN-transformer approach for remote sensing super-resolution," *Remote Sens.*, vol. 16, no. 5, p. 880, Mar. 2024.
[38] L. Jiao et al., "Causal inference meets deep learning: A comprehensive survey," *Research*, vol. 7, p. 0467, Jan. 2024.
[39] H. Schneider, "The emergence of enhanced intelligence in a brain-inspired cognitive architecture," *Frontiers Comput. Neurosci.*, vol. 18, May 2024, Art. no. 1367712.
[40] J. Chomik-Morales, N. Kanwisher, L. Schulz, and R. Pramod, "Using fMRI to explore causal reasoning and abstract relational reasoning in the left lateral prefrontal cortex in the human brain," Open Sci. Framework (OSF), Charlottesville, VA, USA, Feb. 2024, doi: 10.17605/OSF.IO/83X74.
[41] T. Felin and M. Holweg, "Theory is all you need: AI, human cognition, and causal reasoning," *Strategy Sci.*, vol. 9, no. 4, pp. 346–371, Dec. 2024.
[42] J. Pearl," *Causality*. Cambridge, U.K.: Cambridge Univ. Press, 2009.
[43] Y. Wang, L. Meng, H. Ma, Y. Wang, H. Huang, and X. Meng, "Modeling event-level causal representation for video classification," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 3936–3944.
[44] M. Yang, C. Xu, A. Wu, and C. Deng, "A decomposable causal view of compositional zero-shot learning," *IEEE Trans. Multimedia*, vol. 25, pp. 5892–5902, 2023, doi: 10.1109/TMM.2022.3200578.
[45] X. Zhang, P. Cui, R. Xu, L. Zhou, Y. He, and Z. Shen, "Deep stable learning for out-of-distribution generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 5372–5382.
[46] S. S. Kancheti, A. G. Reddy, V. N. Balasubramanian, and A. Sharma, "Matching learned causal effects of neural networks with domain priors," 2021," *arXiv:2111.12490*.
[47] W. Ding, L. Shi, Y. Chi, and D. Zhao, "Seeing is not believing: Robust reinforcement learning against spurious correlation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 66328–66363.
[48] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
[49] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[50] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.
[51] Y. Huang, D. Jiao, X. Huang, T. Tang, and G. Gui, "A hybrid CNN-transformer network for object detection in optical remote sensing images: Integrating local and global feature fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 241–254, 2025.
[52] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. 14th Eur. Conf. Comput. Vis.*, Amsterdam, The Netherlands. Cham, Switzerland: Springer, Oct. 2016, pp. 21–37.
[53] T.-Y. Ross and G. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Conf. Comput. Vis. pattern Recognit.*, Jun. 2017, pp. 2980–2988.
[54] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," in *Proc. Comput. Vis. Pattern Recognit.*Berlin, Germany: Springer, 2018, pp. 1–6.
[55] Q. Zhao et al., "M2Det: A single-shot object detector based on multi-level feature pyramid network," in *Proc. AAAI Conf. Artif. Intell.*, 2018, pp. 9259–9266.
[56] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.
[57] H. Lv, W. Qian, T. Chen, H. Yang, and X. Zhou, "Multiscale feature adaptive fusion for object detection in optical remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[58] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.*Cham, Switzerland: Springer, 2020, pp. 213–229.
[59] Y. Wang, X. Zhang, T. Yang, and J. Sun, "Anchor DETR: Query design for transformer-based detector," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 3, Jun. 2022, pp. 2567–2575.
[60] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.
[61] X. Ma, P. Lv, and Y. Zhong, "QETR: A query-enhanced transformer for remote sensing image object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[62] N. Mo and R. Zhu, "A novel transformer-based object detection method with geometric and object co-occurrence prior knowledge for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 2383–2400, 2025.
[63] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.
[64] Y. Zhang, Y. Yuan, Y. Feng, and X. Lu, "Hierarchical and robust convolutional neural network for very high-resolution remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5535–5548, Aug. 2019.



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